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Maths Assaignment 3

Que-1 Explain partial differential with example

Ans A partial differential equation is a mathematical equation that involves partial derivative of a function with respect to the multiple independent independent variables.

For example, the heat equation, $\frac{\partial u}{dt} = \alpha \nabla^2 u$, describe how temperature (u) changes over time (t) in a substance.

Que-2 If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then prove

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$$

Ans Given

$$u = \log(x^3 + y^3 + z^3 - 3xyz)$$

$$\Rightarrow \frac{\partial u}{\partial x} = \partial\left(\log(x^3 + y^3 + z^3 - 3xyz)\right)$$

Here $\frac{d}{dx}(\log x) = \frac{1}{x}$, $x > 0$

$\frac{d}{dx}(x^n) = nx^{n-1}$ and $\frac{d}{dx}(\text{constant}) = 0$

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$$\Rightarrow \frac{\partial u}{\partial x} = \frac{1}{\log(x^3+y^3+z^3-3xyz)} \cdot (3x^2+0+0-3xyz)$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{3x^2-3xyz}{\log(x^3+y^3+z^3-3xyz)} \quad -(1)$$

$\Rightarrow$ Similarly,

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{3y^2-3xz}{\log(x^3+y^3+z^3-3xyz)} \quad -(2)$$

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{3z^2-3xy}{\log(x^3+y^3+z^3-3xyz)} \quad -(3)$$

Now,

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3x^2-3xyz+3y^2-3x^2+3z^2-3xy}{x^3+y^3+z^3-3xyz}$$

$$\frac{3(x^2+y^2+z^2-xy+yz+xz)}{x^3+y^3+z^3-3xyz}$$

$$\frac{3(x^2+y^2+z^2-xy+yz+xz)}{(x+y+z)(x^2+y^2+z^2-xy+yz+xz)}$$

$\therefore x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$

$$\Rightarrow \frac{\partial u}{dx} + \frac{\partial u}{dy} + \frac{\partial u}{dz} = \frac{3}{(x+y+z)}$$

Hence Proved

Que-3 If $y = \sin^{-1} \dfrac{x^2+y^2}{x+y}$, then prove that

$$x \cdot \frac{\partial t}{\partial x} + y \frac{\partial u}{\partial y} = \tan(u)$$

Solve $\sin u = \dfrac{x^2+y^2}{x+y}$

differentiating with respect to x

we get, $\cos u \dfrac{du}{dx} = \dfrac{2x(x+y)-(x^2+y^2)}{(x+y)^2}$

$$\cos u \frac{du}{dx} = \frac{2y(x+y)-x^2+y^2}{(x+y)}$$

$$= x \cdot \frac{dy}{dx} + y\frac{du}{dy} - \sin u \left(\frac{y^3+y^3+x}{(x+y)}\right)$$

$$= \sec u \cdot \sin u$$

$$= \tan(u)$$

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Que 4 If f is a homogenous function of degree n, then.

$$y \frac{\partial^2 f}{\partial y^2} + x \frac{\partial^2 f}{\partial x \partial y} = (n-1) \frac{\partial f}{\partial y}$$

Solve let (By euler's thearam)

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf \quad \text{--- (1)}$$

By eq (1) the relative Partial differtition of x, then,

$$\frac{\partial f}{\partial x} + x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{n \partial f}{\partial x}$$

$$\Rightarrow x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} = (n-1) \frac{\partial f}{\partial x}$$

Again in eq (1) the relative Partial differetiation of y, then

$$x \cdot \frac{\partial^2 f}{\partial y \partial x} + \frac{\partial f}{\partial y} + y \frac{\partial^2 f}{\partial y^2} = n \frac{\partial f}{\partial y}$$

$$= x \cdot \frac{\partial^2 f}{\partial y \partial x} + y \frac{\partial^2 f}{\partial x \partial y} = (n-1) \frac{\partial f}{\partial y}$$

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Ques If f is a hommogeneous function of degree n. then

$$x^2 \frac{d^2 f}{dx^2} + y^2 \frac{\partial^2 f}{\partial y^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} = n(n-1) f$$

Ans let by eular's law

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf \quad \text{---(1)}$$

by eq (1) the relative partial differentiation of x, then,

$$\frac{\partial f}{\partial x} + x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} = \frac{n \partial f}{\partial x}$$

$$\Rightarrow x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} = (n-1) \frac{\partial f}{\partial x}$$

Again the eq (1) the relative partial differ. of y then,

$$x \frac{\partial^2 f}{\partial y \partial x} + \frac{\partial f}{\partial y} + y \frac{d^2 f}{\partial y^2} = n \frac{\partial f}{\partial y}$$

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$$\Rightarrow x \frac{\partial^2 y}{\partial y \partial x} + y \frac{\partial^2 f}{\partial y^2} = (n-1) \frac{\partial f}{\partial y}$$

Now eq (2) multiply by x and eq (3) multiply by y and then sum of both, we get,

$$x^2 \frac{\partial^2 f}{\partial x^2} + xy \frac{\partial^2 f}{\partial x \partial y} + xy \frac{\partial^2 f}{\partial y \partial x} + y^2 \frac{\partial^2 f}{\partial y^2}$$

$$= (n-1) x \frac{\partial f}{\partial x} + (n-1) y \frac{\partial f}{\partial y}$$

$$\Rightarrow x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 y}{\partial y^2} =$$

$$(n-1) \left[x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right]$$

$$\left\{ \because \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} \right\}$$

$$x^2 \cdot \frac{\partial^2 f}{\partial x^2} + y^2 \frac{\partial^2 f}{\partial y^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} = n(n-1) f$$

from eq (1)

Hence proved Ans